

Hafiz Abdul Rehman

Data Scientist

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Abdul-Insighht

Education

- 2022 – Present **Bachelor of Science in Data Science**, *Gift University*, Gujranwala, Pakistan, CGPA: 3.6/4.0
- 2020 – 2022 **FSC Pre-Engineering**, *ACE Group of Colleges*, Grade: A
- 2018 – 2020 **Matric in Science**, *Iqra Huffaz Secondary School*, Grade: A+

Professional Experience

Dec 2022 – Present **Senior Data Scientist & ML Engineer**

- Developed and deployed advanced machine learning & DL models to enhance personalized marketing and operational efficiency
- Conducted deep analysis of large datasets using Python and SQL to uncover key business insights
- Built and maintained end-to-end ML pipelines, improving model scalability, accuracy, and robustness
- Created interactive dashboards with Tableau and Power BI for real-time decision support
- Automated reporting systems to accelerate data preparation and improve accuracy
- Collaborated with cross-functional and engineering teams to optimize data pipelines and deliver data-driven solutions
- Produced regular reports influencing strategic business decisions

Technical Skills

- Languages Python, R, SQL, C++, Java, JavaScript, PHP, HTML
- ML/DL Frameworks TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV, MediaPipe
- Data Science Pandas, NumPy, Matplotlib, Seaborn, Plotly
- Tools & Platforms Git/GitHub, Docker, Jupyter, Google Colab, Streamlit, Flask, Tableau, Power BI
- Databases MySQL, PostgreSQL, MongoDB
- Specialized Areas Generative AI, Agentic AI, Computer Vision, Statistical Analysis, Data Visualization, RAG Systems

Research Experience & Skills

- Research Areas Machine Learning, Deep Learning, Foundation Models, Knowledge Distillation, Multi-Modal Learning, Medical Imaging, Computer Vision, Time-Series Analysis, Ensemble Methods, Real-Time Systems
- Technical Strengths Machine Learning Algorithm Development, Deep Learning Architecture Design, Model Optimization & Hyperparameter Tuning, Transfer Learning & Fine-Tuning, Ensemble Methods, Feature Engineering, Real-Time Video Processing, Statistical Analysis
- Soft Skills Independent Research, Technical Writing, Problem Solving, Algorithm Design, Code Documentation, Collaborative Development

Industry Experience

Aug 2025 – **Agentic AI & Machine Learning Engineer (Intern), Nexl Tech**

- Present
- Worked on Agentic AI systems including autonomous task planning, tool usage, and multi-agent workflows
 - Developed and fine-tuned machine learning models for real-world business use cases
 - Implemented Retrieval-Augmented Generation (RAG) pipelines for intelligent document and chatbot systems
 - Built end-to-end AI solutions using Python, LangChain, vector databases, and LLM APIs
 - Collaborated with senior engineers to deploy scalable AI applications with production-ready architecture

Research & Projects

Deep Learning & Medical Imaging

Brain Tumor Segmentation using Attention U-Net, *Advanced Medical Imaging*

- Developed attention U-Net model for 3D brain tumor segmentation on BraTS 2020 MRI dataset, achieving 99.64% validation accuracy and 0.986 IoU
- Engineered multi-modal processing pipeline handling FLAIR, T1CE, and T2 MRI sequences with CLAHE enhancement for improved contrast
- Implemented interactive Streamlit application with 3D visualization using VTK/PyVista for real-time tumor analysis
- Created comprehensive data preprocessing pipeline with normalization, cropping, and automated ROI extraction for 240x240x155 voxel volumes

Kidney Disease Classification System, *Medical Deep Learning*

- Built end-to-end deep learning pipeline for kidney disease classification from CT scan images with production-ready deployment
- Implemented MLOps workflow using DVC for version control, experiment tracking, and model registry management
- Developed Docker containerization strategy for AWS EC2 deployment with CI/CD pipeline using GitHub Actions
- Integrated MLflow for comprehensive experiment tracking, model versioning, and performance monitoring across multiple iterations

COVID-19 Detection using CNN Ensemble, *Ensemble Deep Learning*

- Proposed fuzzy integral-based CNN ensemble combining VGG-11, GoogLeNet, SqueezeNet, and Wide ResNet-50-2 for COVID-19 detection from lung CT scans
- Achieved 98.93% accuracy through advanced ensemble fusion techniques and multi-model aggregation strategies
- Implemented robust preprocessing pipeline with data augmentation and normalization for improved model generalization

Computer Vision & Real-Time Systems

Real-Time Football Player & Team Tracking System, *Sports Analytics*

- Engineered real-time football player detection and team classification system using YOLO object detection and ByteTrack for persistent tracking
- Implemented automatic team classification using Roboflow Sports Library with visual annotation using bounding boxes, ellipses, and triangles
- Developed comprehensive tracking pipeline maintaining unique player IDs across frames for performance analytics
- Created annotated video generation system highlighting players, teams, ball, and referees with color-coded visual elements

Vehicle Speed Detection System, *Traffic Monitoring*

- Developed real-time vehicle speed estimation system using YOLO11 detection with perspective transform for accurate speed calculation
- Implemented camera calibration and top-down view mapping to convert pixel movements into real-world distance measurements
- Integrated ByteTrack for robust multi-object tracking across video frames with unique vehicle identification
- Achieved accurate speed estimation by calculating frame-to-frame displacement with configurable speed thresholds for violation detection

Vehicle Detection & Counting System, *Traffic Analysis*

- Built automated vehicle detection and counting system using YOLO with polygon masking for region-specific analysis
- Implemented real-time vehicle classification distinguishing cars, trucks, buses, and motorcycles with counting logic
- Designed spatial filtering using polygon masks to focus detection on specific road segments and traffic lanes
- Created visualization dashboard displaying real-time counts, vehicle trajectories, and traffic flow statistics

License Plate Detection & Recognition, *Computer Vision*

- Developed automated license plate detection system using YOLO for accurate plate localization in various conditions
- Implemented OCR pipeline for text extraction from detected license plates with preprocessing for improved recognition
- Built end-to-end system handling multiple vehicles simultaneously with real-time processing capabilities

Gesture Recognition & Human Activity Analysis

Sign Language to Text & Speech Conversion, *Accessibility Technology*

- Created real-time American Sign Language recognition system using CNN achieving 97% accuracy across 26 alphabets
- Implemented MediaPipe hand landmark detection with skeleton-based feature extraction for background-invariant recognition
- Developed hierarchical classification strategy grouping similar gestures into 8 classes, then using mathematical operations on landmarks for final classification
- Integrated pyttsx3 for text-to-speech conversion enabling real-time dialogue simulation for hearing-impaired communication

Gesto-Electron Gesture Control System, *Human-Computer Interaction*

- Developed touchless computer control system using hand gesture recognition with MediaPipe for cursor movement and click operations
- Implemented real-time gesture detection for mouse control, enabling hands-free computer interaction
- Created intuitive gesture mapping for common computer operations including clicking, scrolling, and navigation

AI Virtual Mouse, *Computer Vision Application*

- Built computer vision-based virtual mouse system using hand tracking for contactless computer control
- Implemented gesture recognition for click, drag, and scroll operations with real-time responsiveness
- Designed calibration system for adaptive performance across different lighting conditions and hand sizes

Human Activity Recognition System, *Action Classification*

- Developed deep learning system for recognizing human activities from video sequences using CNN-LSTM architecture
- Implemented temporal feature extraction capturing motion patterns across consecutive frames
- Created classification pipeline distinguishing multiple activity categories with real-time inference capabilities

Real-Time Gaze & Emotion Tracking, *Behavioral Analysis*

- Built multi-task deep learning system for simultaneous gaze direction and emotion recognition from facial analysis
- Implemented real-time processing pipeline using facial landmark detection and emotion classification CNN
- Developed attention mechanism to improve accuracy in challenging conditions with varying lighting and poses

Interactive Gaming & Creative Applications

Subway Surfer Virtual Controller, *Game Control System*

- Created hand gesture-based controller for Subway Surfer game using MediaPipe hand tracking
- Implemented gesture-to-keyboard mapping enabling touchless game control through hand movements
- Designed real-time gesture recognition with low latency for responsive gaming experience

Neural Networks 3D Digit Counting, *Deep Learning Visualization*

- Developed interactive 3D visualization system demonstrating neural network training for digit recognition
- Implemented real-time training visualization showing weight updates and accuracy progression
- Created educational tool illustrating deep learning concepts with interactive parameter adjustment

Smart Surveillance & Safety Systems

Social Distance Violation Detection System, *Public Safety*

- Built real-time social distancing monitoring system using YOLO person detection with distance calculation
- Implemented perspective transformation for accurate distance measurement between detected individuals
- Created violation alert system with visual indicators highlighting social distancing breaches
- Designed configurable distance thresholds adaptable to different environments and safety regulations

Illegal Parking Detection System, *Smart City Solution*

- Developed automated parking violation detection using computer vision with designated zone monitoring
- Implemented vehicle detection and tracking with dwell time analysis for identifying illegal parking
- Created alert generation system with timestamp logging and violation evidence capture

Line Follower Detection, *Robotics Vision*

- Built computer vision system for autonomous line following robot navigation using edge detection
- Implemented real-time path detection algorithm processing video feed for trajectory calculation
- Created control signal generation for robot steering based on line position and orientation

AI-Powered Applications & RAG Systems

Student Conversational YouTube RAG System, *Educational AI*

- Developed intelligent learning platform using RAG (Retrieval-Augmented Generation) for interactive YouTube content exploration
- Implemented Gemini 2.0 Flash AI integration for natural language understanding and contextual response generation
- Created video transcript extraction and embedding system with semantic search for relevant content retrieval
- Built conversational interface enabling students to ask questions and receive time-stamped answers from educational videos

Research Components Extractor Using RAG, *Academic Tool*

- Built automated research paper analysis system using RAG to extract key components including methodology, results, and conclusions
- Implemented NLP pipeline for document parsing, section identification, and information extraction
- Created structured output generation organizing research findings into standardized academic formats

Barber Salon Chatbot Assistant Using RAG, *Business Automation*

- Developed conversational AI assistant for salon booking and customer service using RAG architecture
- Implemented appointment scheduling, service recommendations, and FAQ handling through natural language interaction
- Created knowledge base integration with real-time availability checking and booking confirmation

Application Tracking System Using RAG, *HR Technology*

- Built intelligent ATS system using RAG for automated resume screening and candidate ranking
- Implemented semantic matching between job descriptions and candidate profiles using embedding models
- Created conversational interface for recruiters to query candidate database using natural language

Calories Advisor ChatBot, *Health & Wellness*

- Developed AI-powered nutrition advisor providing personalized calorie recommendations using conversational interface
- Implemented meal planning suggestions based on dietary preferences, restrictions, and health goals
- Created food database integration with nutritional information retrieval and calorie tracking

AutoStatAgent: Automated Data Analysis System, *Data Science Automation*

- Developed agent-based system for automated statistical analysis and LaTeX report generation
- Implemented intelligent data exploration with automatic visualization selection and statistical test recommendations
- Created end-to-end pipeline from raw data input to publication-ready LaTeX documents

Conversational Chatbot, *NLP Application*

- Built general-purpose conversational AI chatbot with context awareness and multi-turn dialogue handling
- Implemented intent recognition and entity extraction for understanding user queries
- Created response generation system with personality modeling for engaging conversations

Machine Learning & Prediction Systems

Loan Prediction System, *Financial ML*

- Developed machine learning model predicting loan approval likelihood using applicant financial and demographic data
- Implemented feature engineering pipeline extracting relevant predictors including credit history, income, and debt ratios
- Created classification model with hyperparameter optimization achieving high prediction accuracy for risk assessment

Salary Prediction System, *HR Analytics*

- Built regression model predicting employee salaries based on experience, education, location, and job role
- Implemented comprehensive data preprocessing handling categorical variables, missing values, and outliers
- Created feature importance analysis identifying key salary determinants for compensation strategy

Movie Recommendation System, *Collaborative Filtering*

- Developed personalized movie recommendation engine using collaborative filtering and content-based approaches
- Implemented user-item matrix factorization for discovering latent preference patterns
- Created hybrid recommendation system combining multiple algorithms for improved suggestion accuracy

Certifications

- 2025 Agentic AI and AI Agents for Leaders | Coursera
- 2025 Prompt Engineering for ChatGPT | Coursera
- 2025 OpenAI GPTs: Creating Your Own Custom AI Assistants | Coursera
- 2025 Agentic AI and AI Agents: A Primer for Leaders | Coursera
- 2025 Programming with Generative AI (IIT) | Coursera
- 2024 Applied Data Science with Python
- 2024 Data Analysis with R Programming
- 2024 Data Science and Analytics | HP LIFE & DataCamp